“ Mittal Variety Store: “Driving B2C Success Through Data Driven Solutions Stock Optimization and Sales growth. ”

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Organization Background

Mittal Variety Store is an independently owned retail outlet located in Jaito, Punjab, and managed by Mr. Rajinder Kumar for over five years. The store operates under a Business-to-Consumer (B2C) model, offering a wide range of daily essentials and general merchandise. Its operations follow a traditional approach, including manual billing and inventory management with limited digital integration. Despite competition from organized retail chains and e-commerce platforms, the store maintains a strong and loyal local customer base due to its reliability and accessibility.

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Problem Statement

1. Inventory Management Issues

The store faces difficulties in maintaining accurate inventory records, leading to frequent stockouts and overstock situations. Poor inventory control, especially for products with shorter shelf life, increases the risk of losses and negatively impacts profitability.

2. Pricing Strategy Challenges

There is limited knowledge and experience in setting prices that are competitive, reasonable, and profitable. This makes it challenging to attract customers while maintaining sustainable profit margins.

3. Lack of Structured Credit System

The absence of an organized credit management system affects cash flow, resulting in delayed payments and inadequate record-keeping. Implementing structured credit policies is essential to reduce financial risk and ensure timely collections.

Data Overview & Dataset Details

To address the identified business challenges, two structured datasets covering January 2025 to June 2025 were collected from sales bills and inventory records of Mittal Variety Store. The data supports analysis of sales performance, inventory movement, and credit management.

01. Grocery Sales Dataset

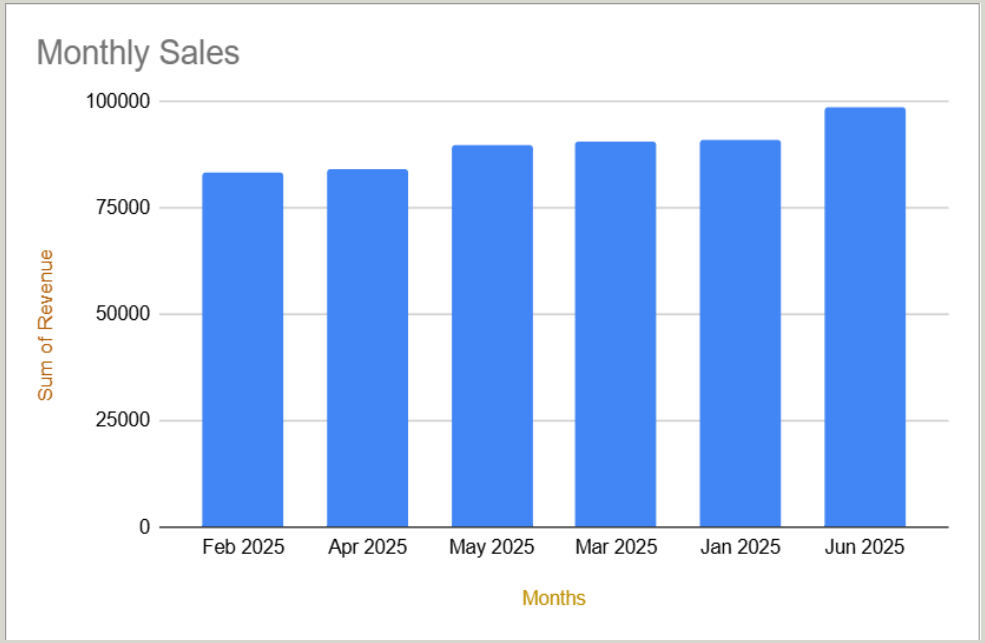
- 4,065 Records | 12 Attributes
- Includes product details, quantity sold, selling price, revenue, profit
- Contains payment mode and credit sales indicators
- Used for analyzing sales trends, profitability, and customer payment behavior

02. Grocery Stocks Dataset

- 6,155 Records | 6 Attributes
- Includes opening stock, stock received, closing stock
- Used for analyzing inventory movement and stock control efficiency

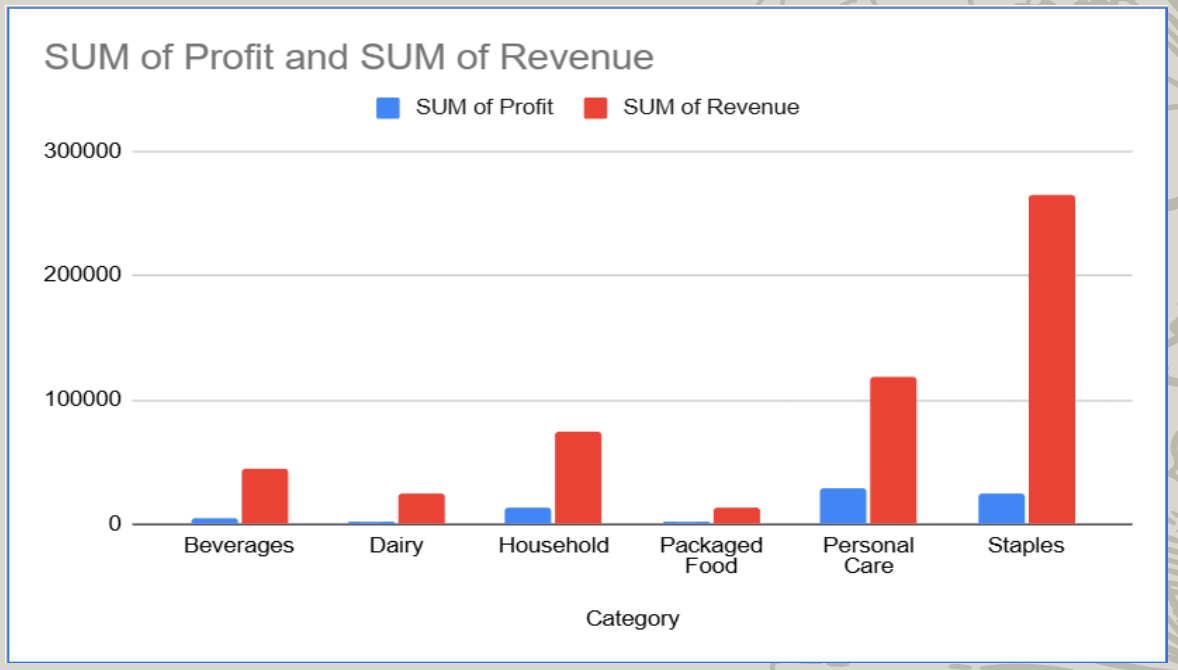
Sales Performance & Revenue Insights

Monthly Sales Trend



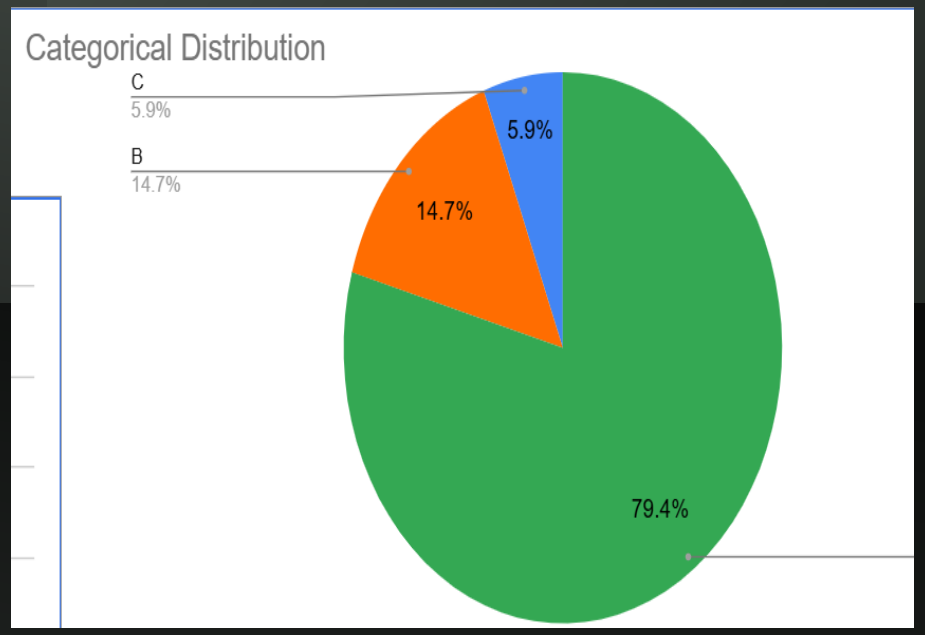
- **Gradually Increasing Trend** over the analysis period.
- Sales are relatively lower during the initial months and **peak in June 2025.**
- Steady revenue growth.

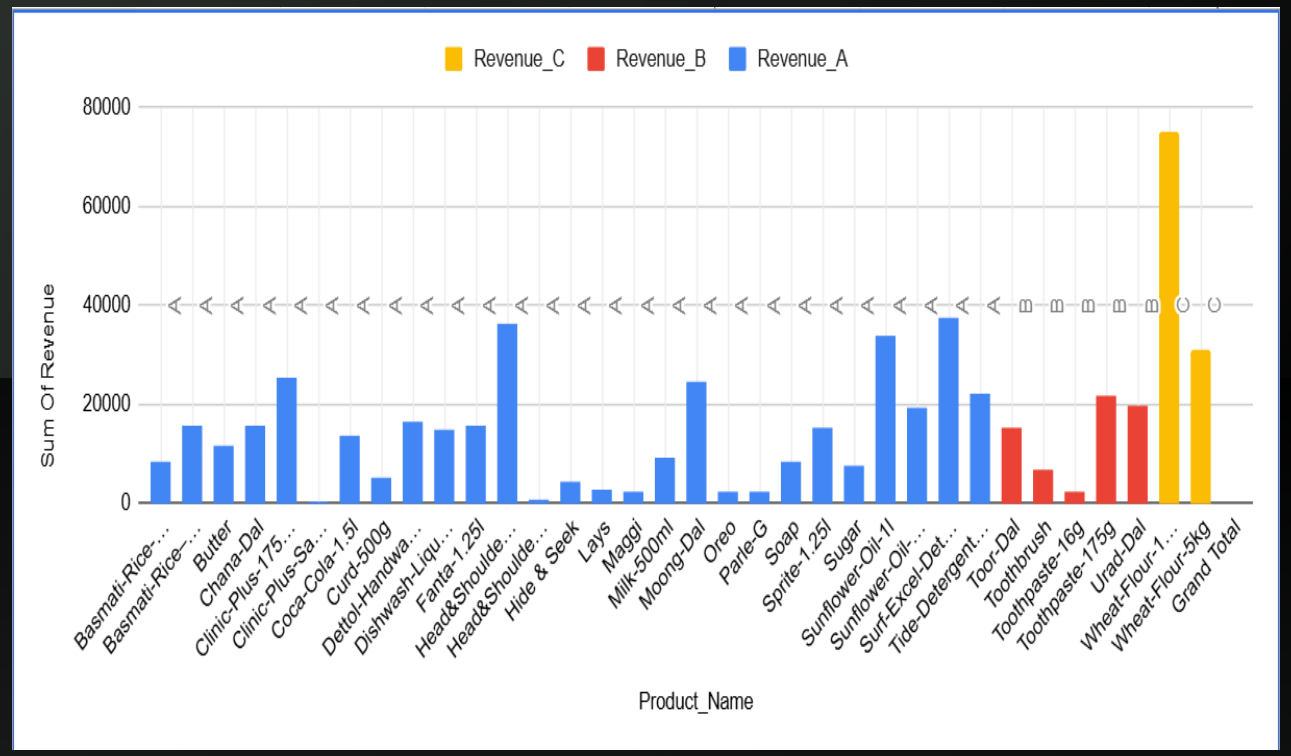
Category wise Revenue vs Profit



- The **Staples** category generates the **highest revenue** as well as profit.
- **Personal Care and Household** also show strong profit contributions relative to revenue.
- Few Categories generate high revenue comparatively low profit.

ABC Analysis & Revenue Concentration

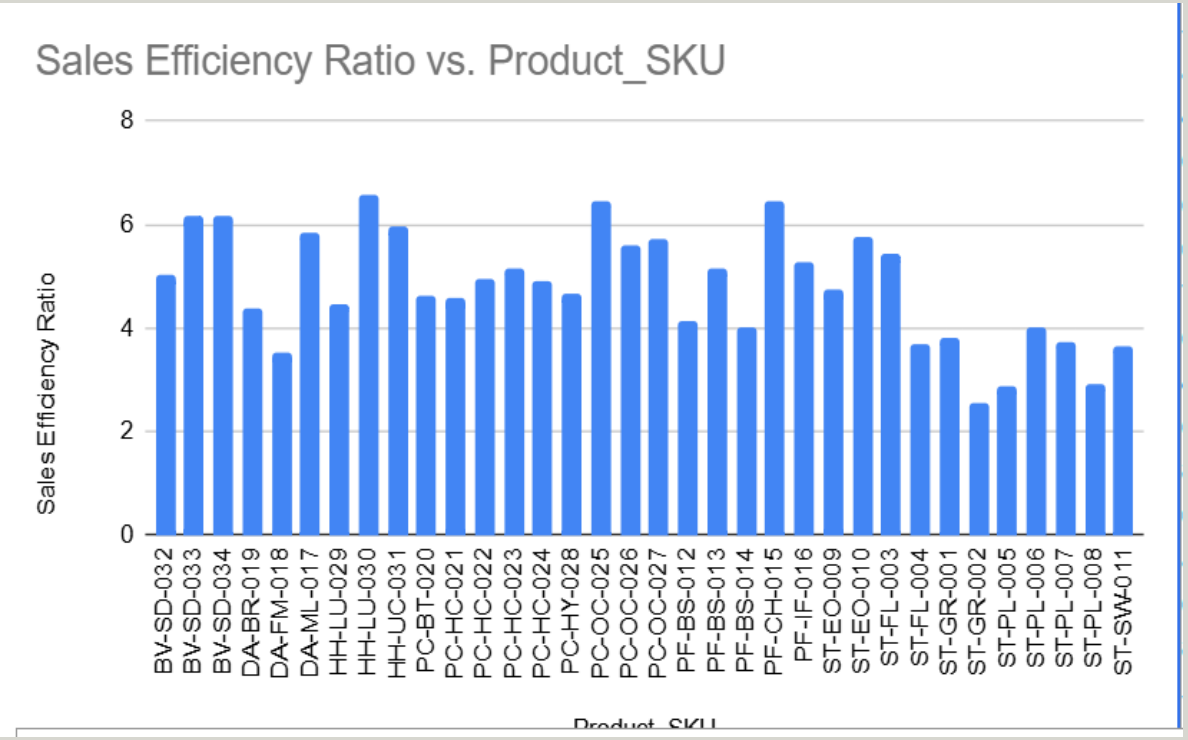


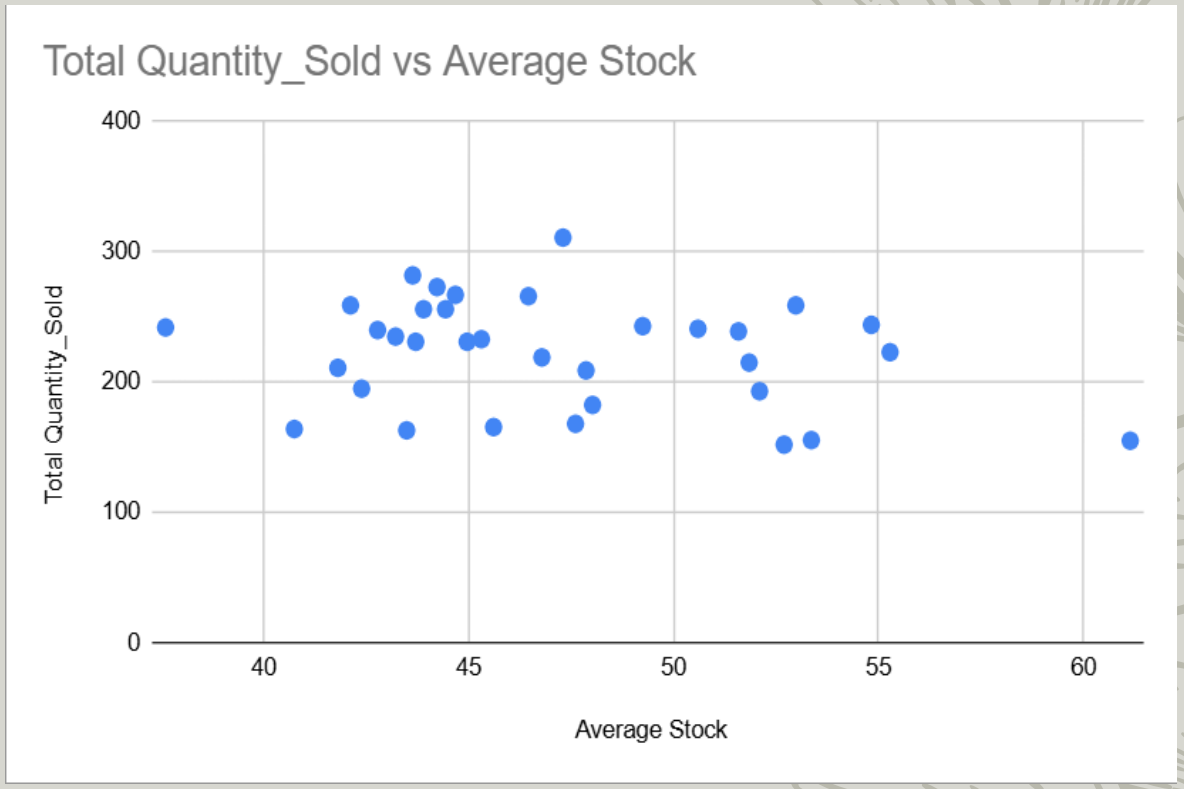


- Follows **Pareto Principle(80-20)** as small number of Products , are contributing to **highest revenue.**
- Category B products account for a moderate share of revenue .
- Category C products contribute the least to overall revenue

- High-revenue essential items like Wheat Flour, Surf Excel, and Sunflower Oil fall under Category A.
- Toor Dal (pulses) and personal care products are classified under Category B

Sales-Stock Efficiency & Inventory Optimization





- Several SKUs exhibit high efficiency ratios (greater than 5), classifying them as fastmoving products.
- Majority of products fall in **medium efficiency range(3-5)** which indicates stable but improvable inventory performance.
- SKU's record efficiency ratio below 3, categorizing as slow-moving products which indicates overstocking or weaker demand.

- **Weak correlation** between average stock levels and quantity sold.
- Importance of demand-driven stocking rather than **volume-based stocking**.

Demand Forecasting

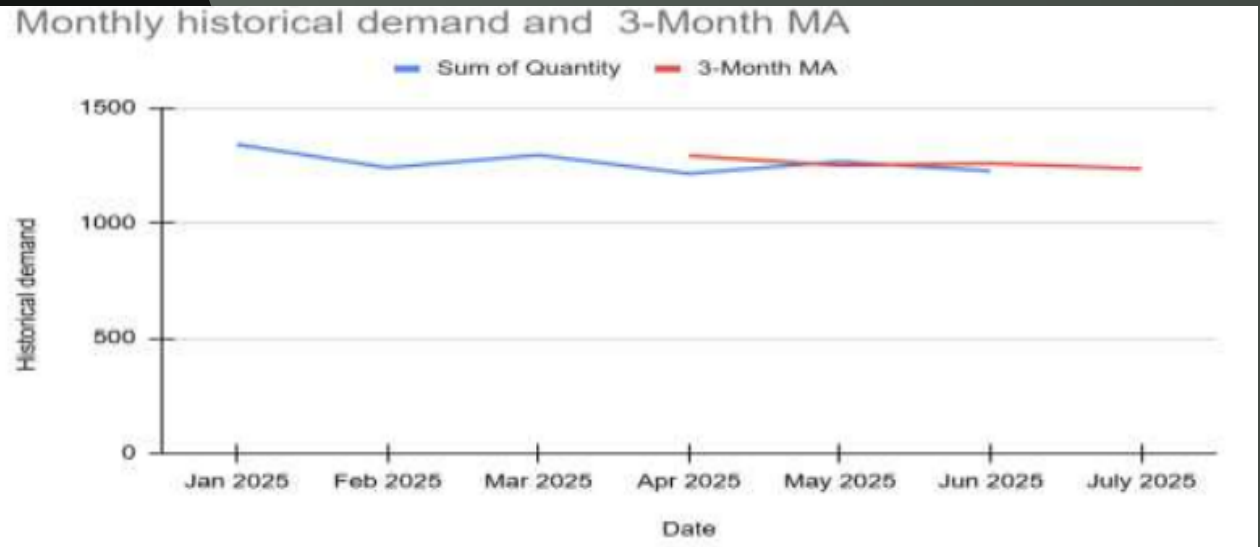
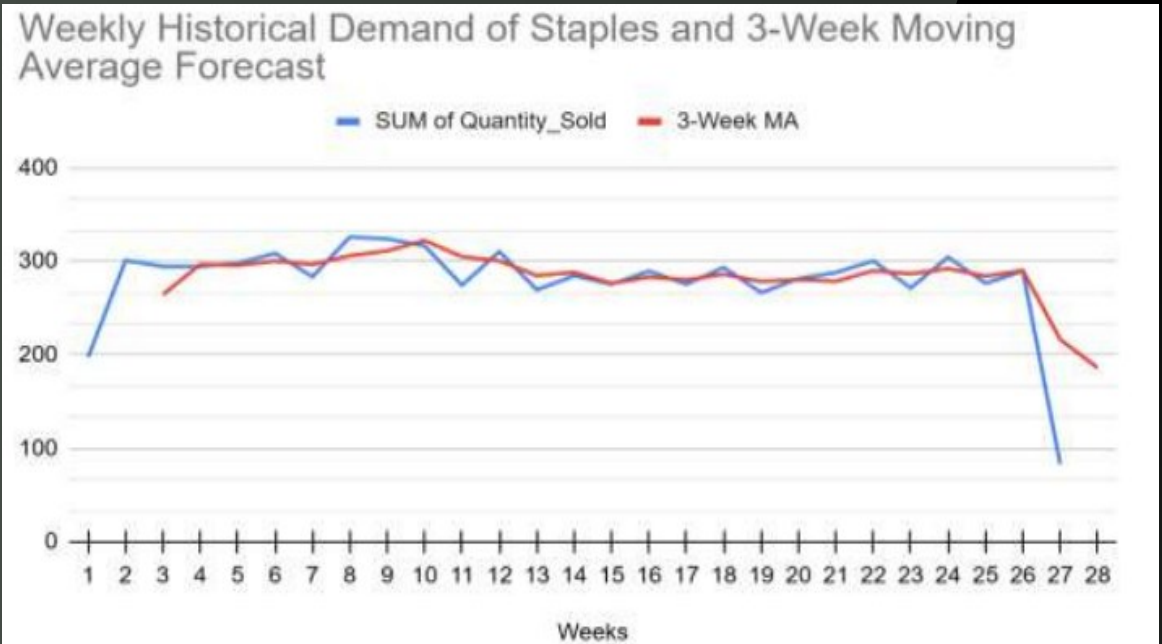


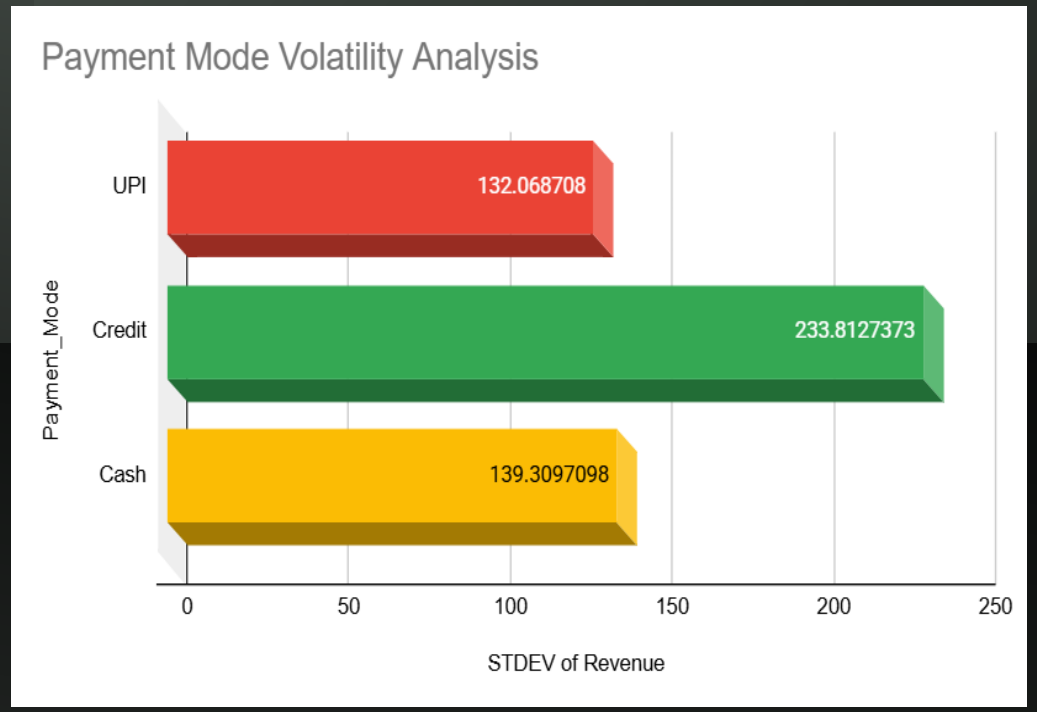
Fig 13: Demand Forecasting of Total sales

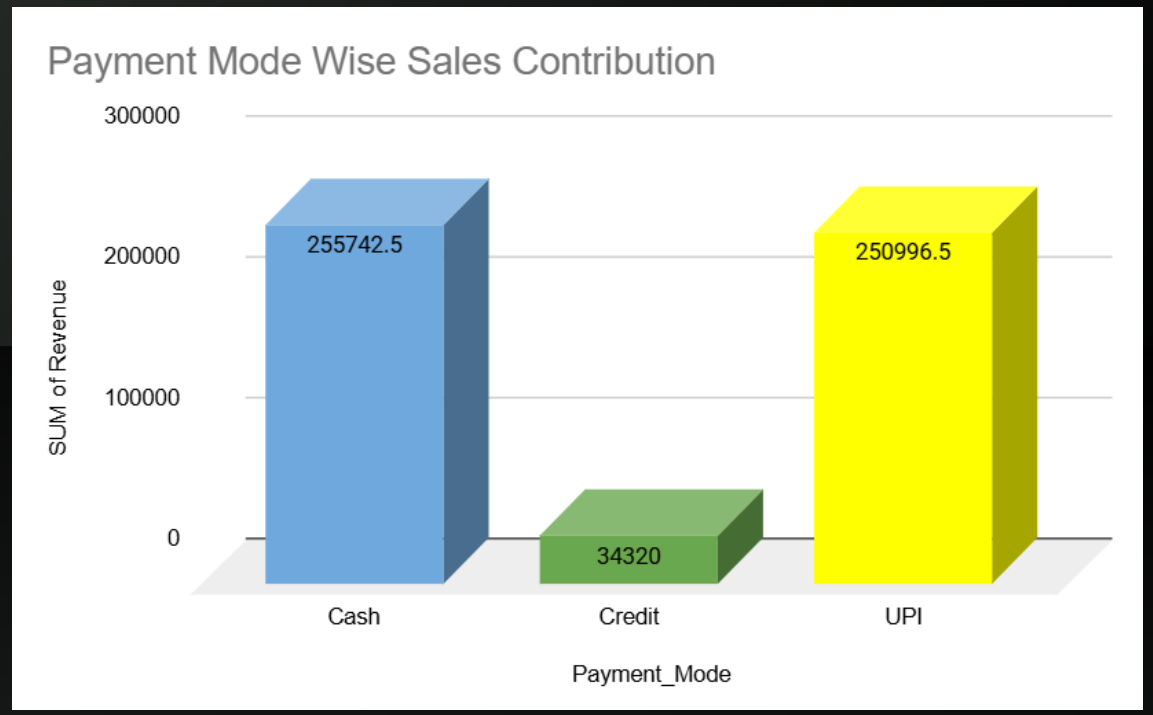


- Demand forecasting guides **Strategic inventory planning.**
- Strong alignment between observed and forecasted demand.
- Monthly demand analysis reveals **gradual demand stabilization.**

- Staples category exhibits relatively stable weekly demand.
- 3-week moving average closely follows actual demand, indicating low volatility.
- **Minor demand peaks observed around mid-weeks.**

Payment Mode & Credit Risk Analysis





- Credit payments exhibit the **highest revenue volatility.**
- Cash and UPI show lower and more stable volatility, indicating predictable daily cash inflows.
- Credit sales introduce **earnings instability despite low volume**

- **Cash and UPI** contribute nearly equal and dominant shares of total revenue.
- Credit sales form a very small portion of total revenue.
- Business is primarily liquidity-driven, reducing short-term cash flow risk.

Key Findings & Strategic Recommendations

Key Findings

- Revenue is primarily driven by non-credit transactions, with UPI and cash dominating total sales.
- Sales-stock efficiency analysis highlights clear segmentation between fast-, medium-, and slow-moving SKUs, indicating uneven inventory utilization.
- Demand for staple and personal healthcare categories remains stable and forecast-friendly.
- Credit sales contribute lower volume but exhibit higher revenue volatility and financial risk.
- ABC analysis confirms revenue concentration among Category A products (Pareto principle).

Strategic Recommendations

- Rationalize slow-moving Category C products to reduce excess inventory costs.
- Implement forecast-driven replenishment planning to minimize stockouts and overstocking.
- Introduce structured credit policies to control volatility and financial risk.
- Promote digital payments (UPI) to enhance stable and predictable cash flow.
- Monitor sales-stock efficiency ratios regularly for proactive inventory optimization.

~ Thankyou